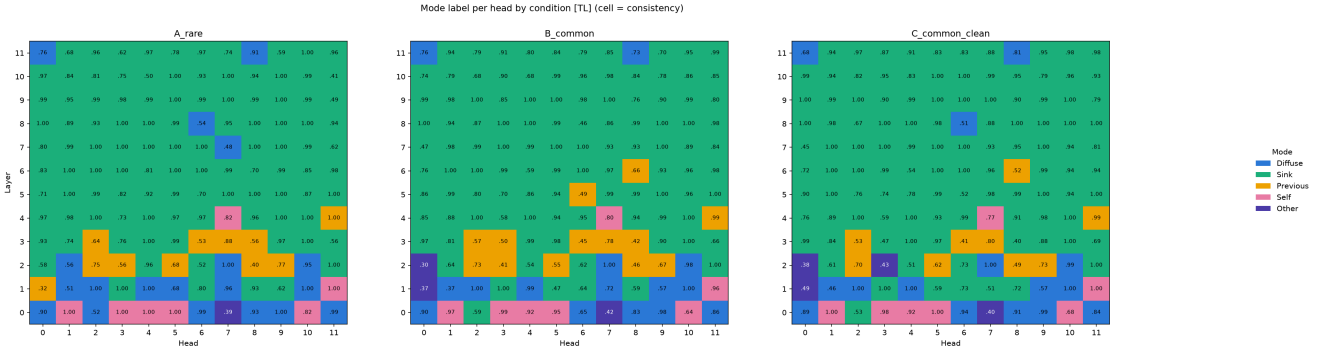
The no-op experiments, one sentence at a time

GPT-2 small · BOS + 32 tokens, attention measured at the final token · “BOS mass” = attention a head parks on the first token (the no-op) · “late” = layers 6–11 · rare / common = the id range of the random filler tokens

Experiment 1 · the no-op baseline (refactored Experiment 1)

`results/20260928-004336_exp1`

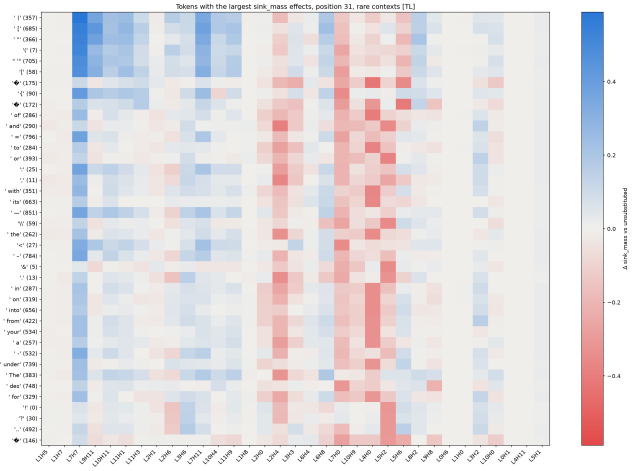
What was tested	Final-token attention of all 144 heads on 1000 random-token rows in three conditions: rare ids, common ids, and common ids without byte tokens.
Hypothesis	Most heads park on BOS on random input, and the token class changes how many do.
Confound	Byte tokens (ids 0–255) could drive heads on their own, so the common condition was run both with and without them.
Results	About 74% of heads park on BOS in every condition, more in late layers, yet 134 of 144 heads shift their BOS mass between rare and common tokens.
Graph	What each head attends to most in the three conditions (green = BOS in this figure), which look almost identical.



Experiment 2 · which tokens move the heads (exp1ext)

`results/20260928-135243_exp1ext`

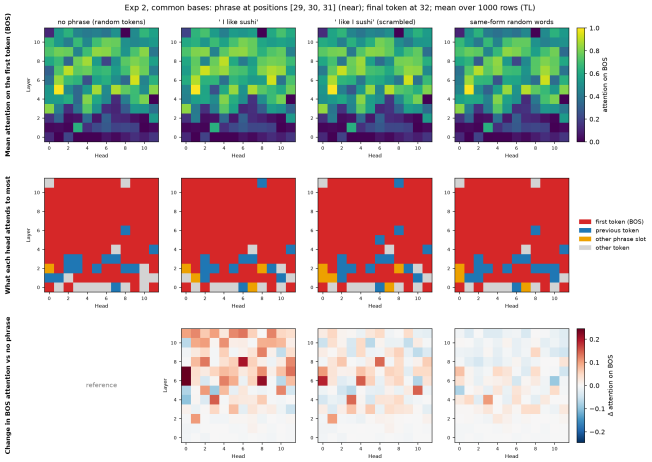
What was tested	A 2×2 of context class × final-token class, plus single tokens swapped into each position.
Hypothesis	The late-head shift comes from the context rather than from the final token.
Confound	Context class and final-token class always changed together in Experiment 1, so the 2×2 crossed them.
Results	The shift came from the whole context, and a single swapped token mattered only at position 31, right before the final token.
Graph	Change in BOS mass per head when one token is swapped in at position 31, by token, with brackets and punctuation moving L7H7 by up to 0.5.



Experiment 3 · a phrase in the noise (Exp 2, iteration 1)

`results/20261003-151225_exp2`

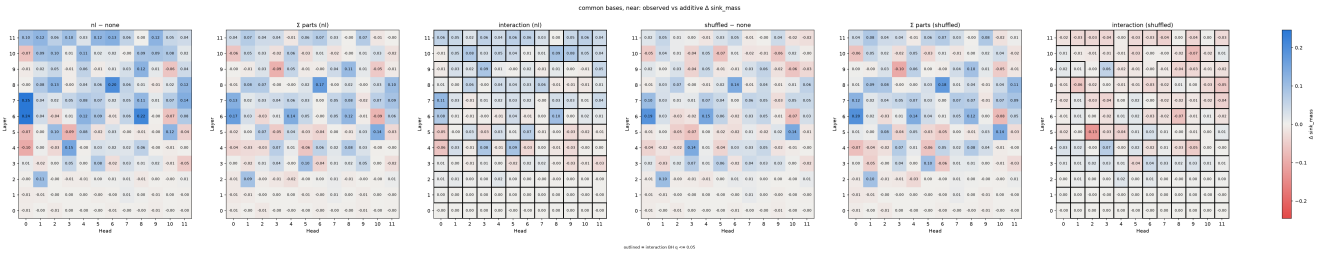
What was tested	“ I like sushi” written at positions 29–31 or 14–16 of random rows, against its scrambled order and same-form random words.
Hypothesis	A coherent phrase right before the final token changes late-head BOS mass more than its controls, and a far placement does little.
Confound	Writing any tokens changes attention, so the phrase was compared with its scrambled order, same-form random words and a second random-word placebo.
Results	Near the final token the phrase raised late BOS mass, word order and the specific tokens mattered, and the far placement was weaker but not null (37 / 25 heads); 3 of 6 predictions held.
Graph	Experiment 1-style maps with the phrase near the final token, where late layers turn red in the change row for the phrase but much less for its controls.



Experiment 4 · more than the sum of its words (Exp 2, iteration 2)

`results/20261003-230841_exp2`

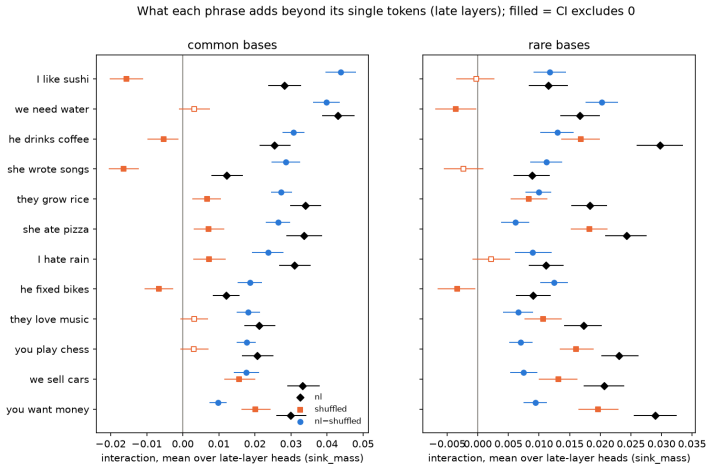
What was tested	Each word of the phrase written alone, plus the model’s surprisal of the words.
Hypothesis	Word effects add up, and the model’s surprise explains the phrase effect.
Confound	A phrase effect could be just its words’ effects or their predictability, so single-word variants and slot surprisal were measured on the same rows.
Results	The phrase exceeded the sum of its words while the scrambled order did not, single words explained none of the order effect ($r^2 \approx 0$), and surprisal explained about a tenth; 7 of 7 predictions held.
Graph	Observed change, sum of single-word changes and their interaction per head, where the phrase’s interaction is positive in late layers.



Experiment 5 · twelve phrases (Exp 2, iteration 3)

results/20261004-125831_exp2

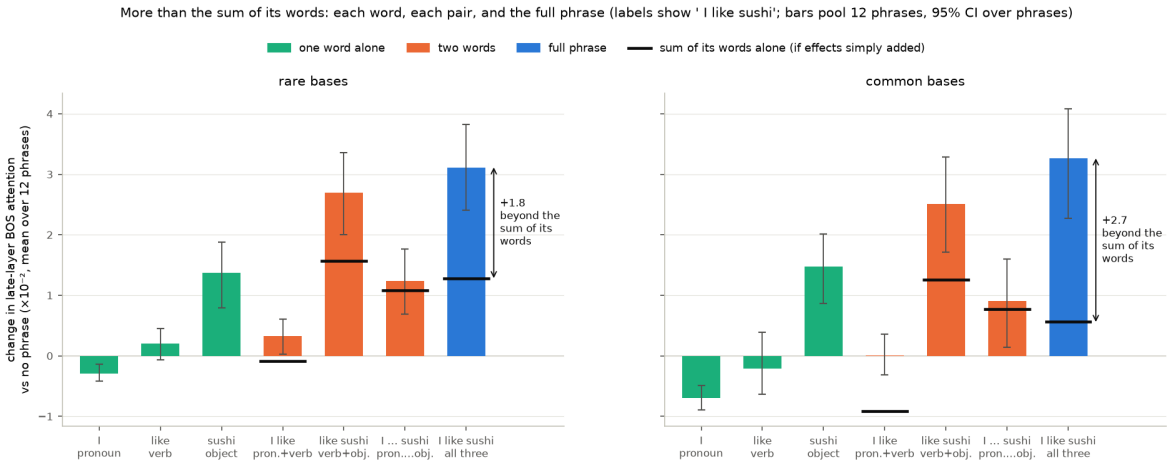
What was tested	Twelve pronoun–verb–object phrases at 29–31, each against its scrambled order and its single words, on both base classes.
Hypothesis	Coherent adjacent words right before the final token make late heads park, for every phrase.
Confound	One phrase could be special and bounded masses can fake interactions, so twelve phrases were tested and the results were repeated on the logit scale.
Results	The phrase-minus-scrambled interaction was positive for 12 of 12 phrases on both bases (+0.025 common, +0.010 rare) and on the logit scale, and surprisal did not explain it; 8 of 10 predictions held.
Graph	Per-phrase interaction terms with CIs, all above zero.



Experiment 6 · which word pairs do the work (Test A)

results/20261004-152303_exp2

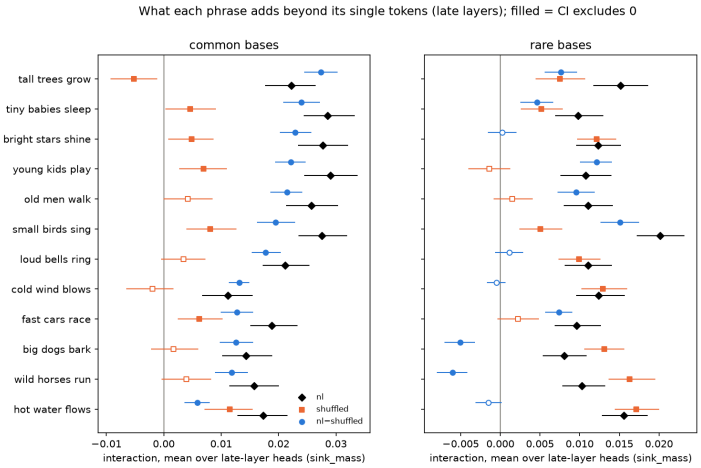
What was tested	Every word pair of the twelve phrases written alone (pronoun–verb, verb–object, pronoun . . . object) against the full phrase.
Hypothesis	The extra BOS mass is mostly carried by the adjacent pair ending right before the final token.
Confound	A three-word effect could come from any pair, so every pair was written alone and the three-way term was separated by decomposition.
Results	The pair ending at 31 carried about half (46% common, 61% rare), the earlier pair a quarter to a third, and the non-adjacent pair and three-way term little; 7 of 8 predictions held.
Graph	Late-layer BOS change for each word, each pair and the full phrase, with ticks at the sum of the words alone.



Experiment 7 · other phrase types (Test C)

results/20261004-155651_exp2

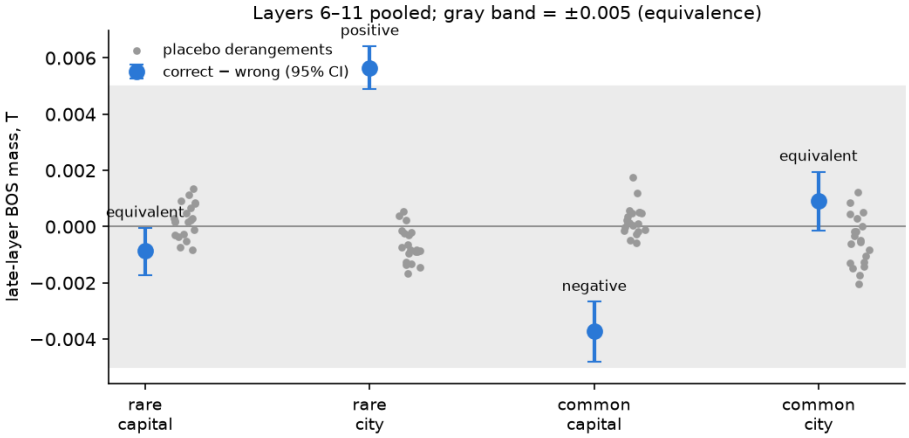
What was tested	Twelve adjective–noun–verb and twelve preposition–determiner–noun phrases with the same controls.
Hypothesis	The phrase effect generalizes beyond pronoun–verb–object grammar.
Confound	The effect could be specific to one template, so two other templates were tested on both base classes with identical controls.
Results	All six template $\times$ base cells showed the effect (pooled CIs above zero), but adjective phrases on rare bases were inconsistent (6 of 12 phrases); 4 of 6 and 6 of 6 predictions held.
Graph	Per-phrase interaction terms for the adjective–noun–verb template, including the weak rare-base cell.



Experiment 8 · a true fact before a random final token (Exp 3, R1-out)

results/20261015-190957_exp3rel

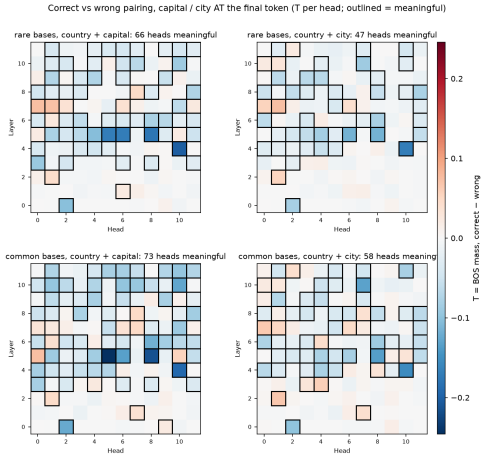
What was tested	Every one of 11 countries paired at positions 30–31 with every capital (and every non-capital city), followed by the row's random final token.
Hypothesis	A true country–capital pair raises late BOS mass more than a false one, as coherent phrases did.
Confound	Truth is tied to which tokens appear, so every country met every capital and each token's own effect cancels in correct $-$ wrong.
Results	For capitals, truth barely changed late BOS mass (-0.0009 rare, -0.0037 common; a correct city raised it only on rare bases, +0.0057), but correct pairs moved attention from the capital onto the country; 1 of 4 predictions held.
Graph	Late-layer correct $-$ wrong BOS mass with 95% CIs against 20 placebo pairings and the ± 0.005 equivalence band.



Experiment 9 · the capital as the final token (Exp 3, R1-in)

results/20261015-203150_exp3rel_in

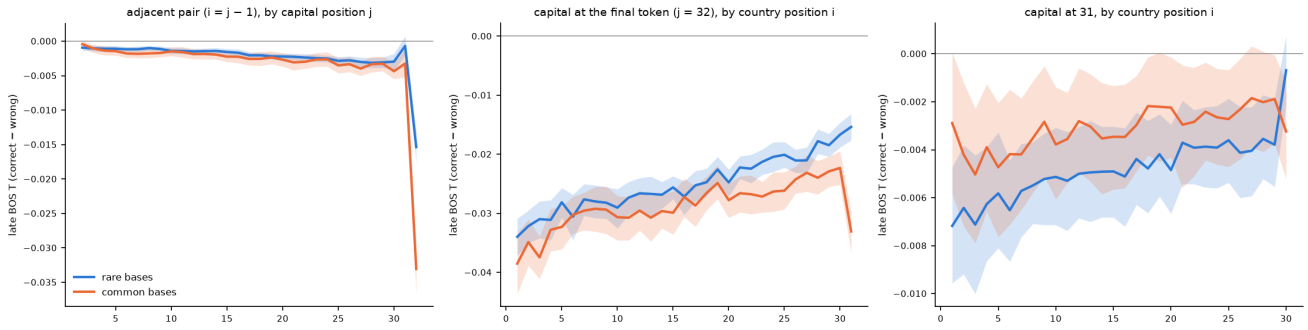
What was tested	The same grids with the country at 31 and the capital or city as the final token, plus a country + random-word anchor.
Hypothesis	A correct capital at the final token takes late heads off BOS and onto the country.
Confound	Changing the final token changes every head's query, so each capital appears once with its own country and ten times with others, cancelling its own effect.
Results	A correct capital cut late BOS mass by 0.018 (rare) and 0.034 (common) in 11 of 11 countries, but most of the freed attention spread over the random context (+0.015 / +0.029) rather than onto the country (+0.002 / +0.006); 3 of 7 predictions held.
Graph	Per-head correct − wrong BOS mass for capitals and cities on both bases, blue across the late layers.



Experiment 10 · every position of the pair (Exp 3, position sweep)

results/20261015-210238_exp3pos

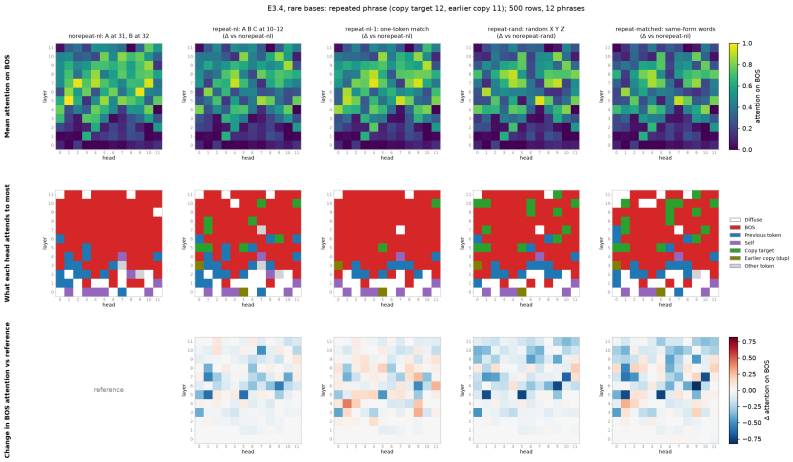
What was tested	The country at every position i and the capital at every later position $j \leq 32$ (496 placements), correct vs wrong pairings.
Hypothesis	The release at the final token survives a gap but weakens with distance, and pairs before a random final token change little.
Confound	Position and distance move together, so all 496 placements used the same rows plus a placebo pairing at every placement.
Results	With the capital as the final token every country position released BOS, on rare bases more strongly the farther away the country was (-0.034 at position 1 vs -0.015 at 31; flat on common), and the placebo was significant at 0 of 496 placements; 3 of 5 predictions held.
Graph	Late BOS change along the adjacent diagonal, with the capital as the final token, and with the capital at 31, by position.



Experiment 11 · copying as a positive control (Exp 3, E3.4)

results/20261015-211838_exp3e34

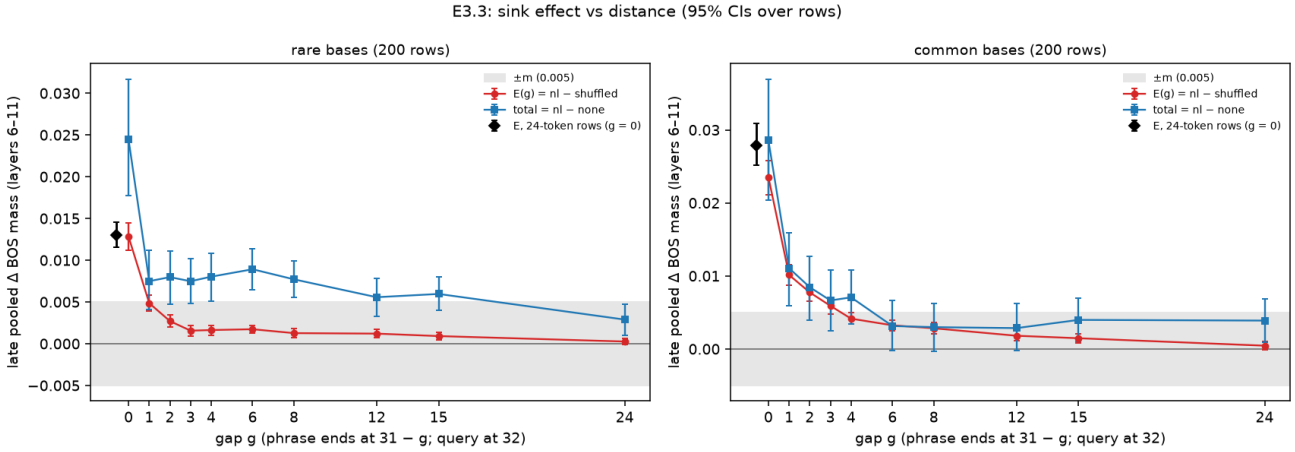
What was tested	A phrase or random block repeated so that the final token's earlier copy is followed by a copy target, against unrepeated controls.
Hypothesis	Known induction heads switch from BOS to the copy target, showing the role metrics can see a head change job.
Confound	Any change in tokens could move attention, so each control differed only at the earlier copy and the final token was identical.
Results	Five to seven heads (L5H0, L5H1, L5H5, L6H9, L7H1, L7H2, L7H10) switched from BOS to the copy target, taking more from BOS than renormalization predicts, while natural phrases were copied less than look-alike words; 3 of 5 predictions held.
Graph	Experiment 1-style maps, where the copy target (green) takes over heads in layers 5–7 on repeats.



Experiment 12 · distance from the final token (Exp 3, E3.3)

results/20261015-214541_exp3e33

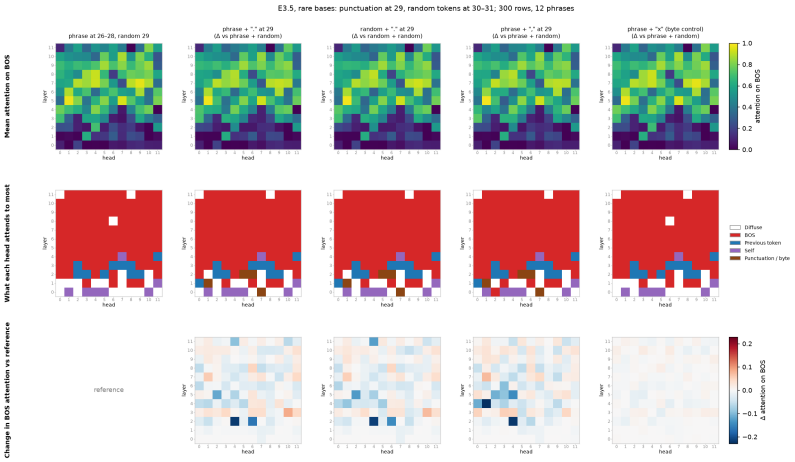
What was tested	The three-word phrase ending g = 0 to 24 tokens before the final token, against its scrambled order and no phrase.
Hypothesis	The phrase effect halves within one token and is gone by a gap of 24.
Confound	Distance is tied to absolute position, so a 24-token-row control kept the gap at zero while moving the phrase to positions 21–23.
Results	Word order mattered mostly next to the final token (rare: +0.013 → +0.005 after one step, a small residual out to 15, zero at 24) while real words still raised BOS 15 tokens away, and absolute position did not matter on rare bases; 3 of 3 predictions held.
Graph	Late BOS change (phrase − scrambled and phrase − none) against the gap g.



Experiment 13 · punctuation as a resting place (Exp 3, E3.5)

`results/20261015-220224_exp3e35`

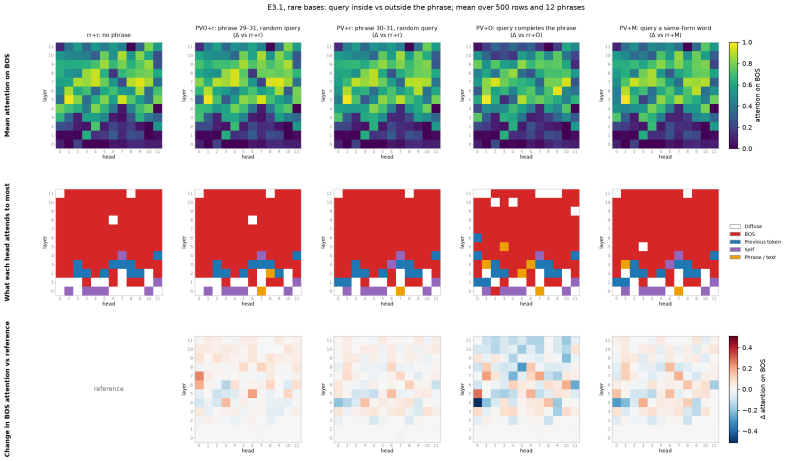
What was tested	A period, comma or newline (or the byte tokens “x”, “7”) three tokens before the final token, after a phrase or after random tokens.
Hypothesis	A period after a phrase becomes an alternative resting place and takes attention from BOS.
Confound	A period is also a rare byte token, so it sat away from the previous-token slot and was compared with the non-punctuation bytes “x” and “7”.
Results	A period drew only +0.007 / +0.013 of late attention and 0–1 late heads switched to it, while early head L2H5 tracked the most recent punctuation; 2 of 3 predictions held.
Graph	Experiment 1-style maps, where the punctuation slot (brown) appears only in early heads such as L2H5.



Experiment 14 · the final token completes the phrase (Exp 3, E3.1)

`results/20261015-221059_exp3e31`

What was tested	The phrase’s first two words at 30–31 with the final token being the completing object, an implausible object, a same-form random word, or a random token.
Hypothesis	When the final token completes the phrase, late heads leave BOS to read it, unlike after an unrelated final token.
Confound	“Completes the phrase” could just mean “the final token is a real word”, so completing objects were compared with same-form words and implausible objects.
Results	The context raised late BOS mass after non-completing final tokens but much less after the completing object (rare: it fell, -0.044; common: +0.008 vs +0.034 for a same-form word), with almost no late heads switching role (1 / 0); 3 of 6 predictions held.
Graph	Experiment 1-style maps, where the change row turns blue for the completing object and red for the random final token.



Experiment 15 · from one word to natural text (Exp 3, E3.2)

results/20261015-223621_exp3e32

What was tested	The last k = 1 to 31 tokens of twelve hand-written passages before a random final token, plus the passage's true next word as the final token.
Hypothesis	Parking saturates by about three coherent tokens, so longer natural text adds nothing.
Confound	More text also means fewer random tokens, so every length was compared with same-form random words and with the scrambled passage.
Results	Late BOS mass kept rising with length to +0.133 (rare) and +0.159 (common) at 31 tokens without needing the final word pair, and the true next word shrank that rise by 0.06; 1 of 4 predictions held.
Graph	Late BOS change against the number of passage tokens, for the passage, its scrambled order and same-form words.

